Drug Repurposing Hypothesis Assessment: Navitoclax for Amyotrophic Lateral Sclerosis

Date: 2026-11-11 · **Task:** Assess, not advocate — evidence against counts as much as evidence for. *Established findings are cited by PMID/NCT with verbatim abstract quotes; interpretive statements are explicitly labeled INFERENCE.*

Summary

The mechanism is real, but the drug is mismatched to it. Cellular senescence with a senescence-associated secretory phenotype (SASP) is a genuine, early, and spatially appropriate feature of ALS. It emerges in the motor cortex and spinal cord of TDP-43 mouse models alongside functional decline (PMID: 42759848), and it is documented in human sporadic-ALS astrocytes, cerebrospinal fluid (CSF), and post-mortem tissue, where it tracks with disease progression (PMID: 42794664). Senolysis is also a biologically validated strategy in ALS-like models — but the only published positive ALS senolytic result used the Dasatinib+Quercetin (D&Q) combination, **not navitoclax** (PMID: 40196013). So the *pathway* the hypothesis names is active and targetable. The problem lies specifically with navitoclax as the chosen tool.

Navitoclax inhibits the two proteins — BCL-2 and BCL-xL — that are protective for the very motor neurons that die in ALS. In the SOD1-G93A model, Bcl-2 overexpression *attenuates* neurodegeneration (PMID: 10582606), and in human ALS tissue Bcl-2 is already depleted in the mitochondrial compartment of vulnerable regions (PMID: 10331434). BCL-xL supports neurite outgrowth and neurotransmission (PMID: 31905614), and a neuroprotective ALS intervention that extended survival did so partly by *raising* Bcl-xL (PMID: 25986728). Navitoclax does the opposite of what these data suggest is protective. Consistent with that tension, a human-neuron Alzheimer's model found navitoclax to be non-selectively neurotoxic while D&Q was safe (PMID: 40713864).

There is no clinical precedent and several structural reasons to doubt the premise. No navitoclax or senolytic trial exists in ALS; the only registered CNS senolytic trials use D&Q in Alzheimer's (NCT04063124, NCT04685590). The "accelerated biological aging" rationale is largely confounded and restricted to a genetic subgroup (PMID: 41025901). ALS is a multistep disease in which single-pathway targeting is likely insufficient (PMID: 41750236), and navitoclax senolysis is itself frequently incomplete owing to MCL-1/GPX4 redundancy (PMID: 39762561). Finally, navitoclax carries dose-limiting, on-target thrombocytopenia (grade 3/4 in 54% of a phase-3 myelofibrosis cohort, PMID: 42378247). **Overall verdict: the case is weak-to-moderate.** The senescence target is credible and worth pursuing, but navitoclax is a poorly suited agent to pursue it with in ALS.

Key Findings

Finding 1 — Cellular senescence is an early, targetable feature in TDP-43 ALS models, and senolysis improves function

Two 2025–2026 studies from the same group establish that molecular markers of cellular senescence emerge early in the motor cortex and spinal cord of TDP-43 ALS mice, coincident with declines in neural and neuromuscular function. The first reports (ESTABLISHED) that *"molecular markers of cellular senescence emerge in the motor cortex and spinal cord alongside declines in neural and neuromuscular function in TDP-43"* (PMID: 42759848). This places the mechanism in exactly the tissues the hypothesis names and at an early, potentially actionable stage.

Crucially, the same group tested a senolytic intervention and reported benefit: *"we investigated the senolytic combination Dasatinib and Quercetin (D&Q) in TDP-43"* (PMID: 40196013), with the paper reporting improved neuromuscular function and preserved cortical excitability. **This is the strongest single piece of support for the broad hypothesis — but it validates senolysis via D&Q, not navitoclax.** These are preclinical (mouse) results; the abstracts report no navitoclax arm and no human tissue quantification.

Finding 2 — Major mechanistic tension: BCL-2/BCL-xL are protective for ALS motor neurons

This is the central objection to the hypothesis. Navitoclax works by inhibiting BCL-2 and BCL-xL to tip primed cells into apoptosis. But in ALS, those anti-apoptotic proteins are *protective* for motor neurons. In the SOD1-G93A familial ALS mouse (ESTABLISHED), *"overexpression of Bcl-2, a protein that inhibits apoptosis, attenuates neurodegeneration produced by the familial ALS-linked SOD1 mutant G93A"* (PMID: 10582606). In human ALS tissue, the protective protein is already depleted where it is most needed: *"the antiapoptotic protein Bcl-2 is decreased in the mitochondrial-enriched membrane compartment of vulnerable regions in ALS"* (PMID: 10331434).

INFERENCE: A drug whose mechanism is to further inhibit BCL-2/BCL-xL is acting in the wrong direction for the neuronal compartment. Even if it selectively kills senescent glia, there is a plausible path to collateral harm in already-vulnerable, BCL-2-depleted motor neurons. This is a liability unique to ALS, absent in the peripheral senolysis indications where navitoclax was developed.

Finding 3 — Navitoclax: CNS senolytic signal in primates, but dose-limiting thrombocytopenia and human-neuron toxicity

The CNS pharmacology is not the limiting problem. In aged nonhuman primates (ESTABLISHED), intermittent navitoclax reached the brain: *"Navitoclax reduced several senescence and SASP biomarkers, with CSF concentrations sufficient for senolysis"* (PMID: 39253182), and reduced TSPO-PET neuroinflammation. However, the same study documents the characteristic on-target toxicity: *"Intermittent treatment with the senolytic navitoclax induced modest reversible thrombocytopenia"* (PMID: 39253182). In humans this toxicity is substantial — a phase-3 myelofibrosis trial reported *"grade 3/4 thrombocytopenia, 54.0% vs 19.2%"* (PMID: 42378247).

Most tellingly, in a human-neuron Alzheimer's model built specifically to test senolytic translatability, navitoclax was toxic to neurons while D&Q was not — the authors conclude navitoclax *"exhibited non-selective neurotoxicity"* (PMID: 40713864). This directly corroborates the mechanistic concern in Finding 2 in a human-cell context.

Finding 4 — Navitoclax reduces neuroinflammatory disease severity in MS/EAE models (cross-disease senolysis)

The best cross-disease support for navitoclax specifically comes from CNS autoimmune neuroinflammation. In experimental autoimmune encephalomyelitis (EAE) (ESTABLISHED), *"middle-aged mice (40-44 weeks) with EAE exhibit improved clinical outcomes and survival following treatment with either dasatinib plus quercetin (D+Q) or navitoclax"* (PMID: 41126823), and human MS data link the target to active lesions: *"in human single-nucleus sequencing from MS, BCL2L1 positive microglia were enriched in lesions with active inflammatory pathology"* (PMID: 39487537).

INFERENCE on transfer: MS/EAE is driven by a senescent *microglial/immune* compartment where killing BCL-xL-dependent cells is beneficial. ALS is a motor-neuron disease where BCL-xL is neuroprotective. The transferability is therefore partial — it supports the senolytic concept in CNS inflammation but does not resolve the ALS-specific neuronal-compartment risk. Notably, the benefit was age-dependent (middle-aged mice), reinforcing a disease-stage constraint.

Finding 5 — Senescence is present in human sporadic ALS astrocytes, CSF, and post-mortem tissue

Human evidence that the pathway is active (not just mouse) comes from a sporadic-ALS study linking purine-metabolism failure to senescence. Loss of adenosine deaminase impairs DNA repair and raises P16 (ESTABLISHED): *"Loss of ADA led to reduced 53BP1-mediated DNA repair and increased P16 levels, which was recapitulated in control iAstrocytes via ADA inhibition"* (PMID: 42794664). Critically, these changes appear in human material and correlate with progression: *"These alterations were recapitulated in SALS CSF and post-mortem tissue, with ageing and sex affecting purine metabolite levels downstream of ADA and positively correlating with disease progression"* (PMID: 42794664). This is the most direct human-tissue evidence that senescence is a real feature of ALS, and it anchors the "mechanism is active" half of the assessment — with the signal strongest in the glial/astrocyte compartment a senolytic should target.

Finding 6 — No navitoclax or senolytic trial exists in ALS; human CNS senolytic experience is limited to D&Q in Alzheimer's

A ClinicalTrials.gov v2 API query (run 2026-11-11) returned zero navitoclax-in-ALS and zero senolytic-in-ALS studies. All registered navitoclax trials are oncology/myelofibrosis (e.g., NCT03222609, NCT01557777). The only registered CNS senolytic trials use D&Q in Alzheimer's: NCT04063124 (*"Senolytic Therapy to Modulate Progression of Alzheimer's Disease"*, completed Phase 1/2) and NCT04685590 (Phase 2, active). **The human senolytic precedent for neurodegeneration uses a different agent (D&Q) and a different disease (AD).** There is no clinical navitoclax neurodegeneration experience at all.

Finding 7 — Raising (not inhibiting) BCL-xL is neuroprotective — reinforces wrong-compartment risk

Independent of the BCL-2 data, BCL-xL specifically is beneficial to neurons (ESTABLISHED): it *"supports neurite outgrowth and neurotransmission by improving mitochondrial function"* (PMID: 31905614). And in the SOD1-G93A ALS model, the neuroprotective agent N-acetyl-L-tryptophan that delayed onset and extended survival worked partly by raising Bcl-xL: *"L-NAT reached the spinal cord, skeletal muscle, and brain... increased Bcl-xL levels"* (PMID: 25986728). Navitoclax's mechanism is the mirror image. This is the second, independent protein (BCL-xL/BCL2L1) for which the ALS literature points to "more is protective," directly contradicting the drug's action.

Finding 8 — Accelerated biological aging in ALS is largely confounded; present only in a genetic subgroup

A key part of the senescence rationale is that ALS reflects accelerated aging. In the largest relevant dataset — Project MinE whole-blood methylomes (5,146 ALS patients, 2,156 controls) — this appears to be an artifact (ESTABLISHED): *"When correcting for cell counts, there is no evidence for accelerated epigenetic aging compared to controls, except for ALS patients with the"* a specific genetic subgroup (PMID: 41025901). This weakens the population-wide rationale and implies that if senescence matters, it may be concentrated in a stratifiable subgroup — with direct implications for trial design (enrich, don't treat-all).

Finding 9 — ALS is a multistep disease; single-pathway senolysis is likely insufficient

Eisen's multistep model frames ALS as 5–6 converging (epi)genetic, environmental, and age-related steps. The model explicitly cautions against single-target approaches (ESTABLISHED): *"Because ALS emerges from the gradual collapse of multiple layers of biological robustness, tackling a single pathway will be insufficient"* (PMID: 41750236). It also situates anti-senescent therapy as most plausible early/preventively: *"Protection against early-life insults, anti-aging, and anti-senescent therapies may curtail step accumulation preventing ALS from exceeding threshold"* (PMID: 41750236). This supports both a redundancy failure mode and a disease-stage/timing constraint — given late, symptomatic diagnosis, monotherapy senolysis may be too little, too late.

Finding 10 — Navitoclax senolysis is heterogeneous and often incomplete (MCL-1 redundancy)

Even granting that senescent cells should be cleared, navitoclax may not clear them. BH3-profiling across diverse senescence models shows (ESTABLISHED) that *"Inhibition of the antioxidant GPX4 or the Bcl-2 family member MCL-1 using small molecule compounds in combination with ABT-263 significantly increased the induction of apoptosis in some, but not all, previously insensitive senescent cells"* (PMID: 39762561) — i.e., many senescent cells resist navitoclax unless MCL-1/GPX4 is co-inhibited. An independent senotherapeutics review lists navitoclax's drawbacks as *"dose-dependent thrombocytopenia, variable therapeutic efficacy, and the emergence of resistance mechanisms"* (PMID: 41807411). Senescent cells co-express BCL-2, BCL-xL, and MCL-1 (PMID: 41298369), so BCL-2/xL blockade alone leaves an escape route. INFERENCE: ALS senescent glia may not be BCL-xL-addicted, so navitoclax monotherapy could fail to clear them even at CNS-adequate exposure.

Mechanistic Model / Interpretation

The hypothesis asks us to accept a chain of reasoning. It holds together at the front end and breaks at the drug-selection step:

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SENESCENCE PRESENT IN ALS?           YES  (Findings 1, 5)
┌──────────────────────────────────────────────────────────┐
│ Senescent glia/neurons accumulate in motor cortex &      │
│ spinal cord; SASP amplifies neuroinflammation.           │
│ Confirmed in TDP-43 mice (<a href="https://pubmed.ncbi.nlm.nih.gov/42759848/" rel="noopener
│ sporadic-ALS astrocytes/CSF/tissue (<a href="https://pubmed.ncbi.nlm.nih.gov/42794664/" rel=
└──────────────────────────────────────────────────────────┘
                   │
                   ▼
DOES CLEARING SENESCENT CELLS HELP?   YES, but via D&Q (Finding 1)
┌──────────────────────────────────────────────────────────┐
│ D&Q improves neuromuscular function & cortical           │
│ excitability in TDP-43 ALS mice (<a href="https://pubmed.ncbi.nlm.nih.gov/40196013/" rel="no
│ Senolysis helps in EAE/MS too (<a href="https://pubmed.ncbi.nlm.nih.gov/41126823/" rel="noop
└──────────────────────────────────────────────────────────┘
                   │
                   ▼
IS NAVITOCLAX THE RIGHT TOOL?        NO  (Findings 2, 3, 7, 10)
┌──────────────────────────────────────────────────────────┐
│ Navitoclax = BCL-2 + BCL-xL inhibitor.                   │
│ BUT in ALS those proteins are NEUROPROTECTIVE:           │
│   • Bcl-2 overexpression attenuates degeneration         │
│     (<a href="https://pubmed.ncbi.nlm.nih.gov/10582606/" rel="noopener noreferrer" title="Vi
│     vulnerable regions (<a href="https://pubmed.ncbi.nlm.nih.gov/10331434/" rel="noopener no
│   • Raising Bcl-xL is neuroprotective (<a href="https://pubmed.ncbi.nlm.nih.gov/31905614/" r
│     25986728).                                           │
│   • Human neurons: navitoclax non-selectively toxic,     │
│     D&Q safe (<a href="https://pubmed.ncbi.nlm.nih.gov/40713864/" rel="noopener noreferrer"
│   • Senolysis often incomplete: MCL-1 redundancy         │
│     (<a href="https://pubmed.ncbi.nlm.nih.gov/39762561/" rel="noopener noreferrer" title="Vi
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The decisive asymmetry: **the hypothesis correctly identifies the target pathway (cellular senescence/SASP) but selects a drug whose mechanism directly opposes a well-documented neuroprotective axis in ALS.** D&Q and navitoclax both achieve senolysis, but by different routes — D&Q via tyrosine-kinase/AKT inhibition, navitoclax via BCL-2/xL inhibition. In a disease where BCL-2/xL protect the dying cell type, the route matters enormously. The existing ALS-model success belongs to D&Q, and the one head-to-head human-neuron comparison favors D&Q on safety.

Comparison of senolytic approaches in the ALS context

Dimension	Navitoclax (BCL-2/xL)	D&Q (kinase/AKT)
ALS-model efficacy	None reported	Improves neuromuscular function, preserves cortical excitability (P 40196013)
CNS senolytic signal	Yes — primate CSF (P 39253182)	Human AD trials registered (NCT04063124/04685590)
Human-neuron safety	Non-selective neurotoxicity (P 40713864)	Safe in same model (P 40713864)
Target vs. ALS biology	Opposes neuroprotective BCL-2/xL (P 10582606 P 31905614 P 25986728)	Not directly opposed
Systemic toxicity	Grade 3/4 thrombocytopenia 54% (P 42378247)	Milder, no on-target platelet effect
Completeness of senolysis	Often incomplete, MCL-1 escape (P 39762561)	Variable, different escape profile

Where the hypothesis is strongest vs. weakest

Hypothesis sub-claim	Verdict	Basis
Senescence/SASP active in ALS motor cortex & cord	**Supported**	P 42759848 P 42794664
Senescence confined to a subgroup, biomarker-identifiable	**Partially supported**	Accelerated aging only in genetic subgroup (P 41025901)
Senolysis benefits ALS features	**Supported (D&Q only)**	P 40196013
Navitoclax *specifically* would benefit ALS	**Not supported / contraindicated**	P 10582606 P 31905614 P 40713864
Single-pathway senolysis sufficient	**Unlikely**	P 41750236 P 39762561

Evidence Base

PMID / ID	Study (abbrev.)	Role in assessment
42759848	Senescence early in ALS TDP-43	**Supports mechanism** — senescence in motor cortex/cord
42794664	ADA/purine dysfunction → senescence in sporadic ALS	**Supports mechanism** — human CSF/ tissue, tracks progression
40196013	Senolytic (D&Q) in TDP-43 ALS	**Supports senolysis** — but D&Q, not navitoclax
10582606	Bcl-2 overexpression attenuates SOD1-G93A	**Against drug** — BCL-2 is protective
10331434	Bcl-2 decreased in human ALS vulnerable regions	**Against drug** — target already depleted
31905614	Bcl-xL supports neurite outgrowth/neurotransmission	**Against drug** — BCL-xL beneficial to neurons
25986728	L-NAT extends ALS survival by raising Bcl-xL	**Against drug** — raising, not lowering, Bcl-xL helps
40713864	Human-neuron AD model	**Against drug** — navitoclax neurotoxic, D&Q safe
39253182	Navitoclax in aged primates	**Mixed** — CNS senolysis achievable; thrombocytopenia
42378247	TRANSFORM-1 phase 3 myelofibrosis	**Safety** — grade 3/4 thrombocytopenia 54%
41126823	Senolytics in EAE/MS	**Cross-disease support** — navitoclax improves CNS neuroinflammation
39487537	BCL2L1+ microglia in MS lesions	**Cross-disease support** — target in active inflammation
41025901	Epigenetic age in ALS (n=5,146)	**Weakens rationale** — accelerated aging confounded
41750236	ALS multistep model	**Weakens rationale** — single pathway insufficient; timing
39762561		

PMID / ID	Study (abbrev.)	Role in assessment
	Apoptotic priming / BH3 profiling	**Weakens drug** — MCL-1 redundancy, incomplete senolysis
41807411	Senotherapeutics review	**Weakens drug** — variable efficacy, resistance
41298369	MCL-1 inhibition to clear senescent cells	**Context** — senescent cells co-express BCL-2/xL/MCL-1
NCT04063124 / NCT04685590	D&Q in Alzheimer's	**Precedent** — only CNS senolytic trials; not navitoclax, not ALS

Limitations and Knowledge Gaps

1. **No ALS-specific navitoclax data exist** — neither in vivo efficacy nor human exposure. The entire assessment of navitoclax-in-ALS is therefore inferential, built from (a) senolysis evidence in ALS models using *other* agents, (b) navitoclax evidence in *other* CNS diseases, and (c) BCL-2/xL biology in ALS. This is clearly labeled inference, not established fact.
2. **Compartment selectivity is unresolved.** The core objection assumes that inhibiting BCL-2/xL cannot be made selective enough to spare neurons while killing senescent glia. If senescent glia are far more apoptotically "primed" than motor neurons, a therapeutic window might exist. No ALS study has measured apoptotic priming (BH3 profiling) across ALS cell types, so the window's existence is unknown.
3. **The two foundational ALS apoptosis papers (** P 10582606 P 10331434 **) are from ~1999–2000** and are SOD1/post-mortem based. SOD1-ALS is a minority of cases; TDP-43 biology dominates. Whether BCL-2/xL are equally protective in TDP-43-driven neurons has not been directly re-tested with modern methods.

4. **The strongest pro-senolysis ALS result (** P 40196013 **) and the senescence-presence results (** P 42759848 **) come from a single group** and are recent (2025–2026); independent replication is pending.
5. **Subgroup identification is immature.** The accelerated-aging signal was confined to an unspecified genetic subgroup (P 41025901). We do not yet have a validated biomarker (e.g., CSF p16, SASP cytokine panel, TSPO-PET) prospectively linked to senescence burden in living ALS patients to enrich a trial.
6. **Thrombocytopenia risk in ALS specifically is unquantified.** ALS patients are often older, may have respiratory compromise, and dysphagia complicates oral dosing; the net safety balance in this population is unknown.
7. **Literature-only assessment; no primary data were analyzed, and PubMed abstract access was intermittent**, so some relevant studies may be missed.

Proposed Follow-up Experiments / Actions

The single most informative next step is NOT a navitoclax trial. Priority order:

1. **BH3 apoptotic-priming profiling across ALS cell types (highest value, lowest cost).** Using iPSC-derived ALS motor neurons, astrocytes, and microglia (and/or post-mortem tissue), measure mitochondrial apoptotic priming and dependence on BCL-2 vs. BCL-xL vs. MCL-1. This directly tests whether a navitoclax therapeutic window exists — i.e., whether senescent glia are selectively vulnerable while motor neurons are protected. Extends the method of P 39762561 to ALS. A negative result (neurons as primed as senescent glia) would effectively close the navitoclax hypothesis.

2. **Head-to-head senolytic comparison in a TDP-43 ALS model.** Navitoclax vs. D&Q vs. D&Q+MCL-1 inhibitor, with motor-neuron counts, neuromuscular function, cortical excitability, SASP markers, and explicit safety/neuronal-loss readouts. This would reveal whether navitoclax is merely inferior or actively harmful relative to the validated D&Q benchmark
(P 40196013).
(P 40713864)
3. **Biomarker development for patient stratification.** Prospectively link a CSF/blood senescence-SASP panel (p16, SASP cytokines) and TSPO-PET neuroinflammation (as in P 39253182) to ALS progression rate. This builds the enrichment biomarker needed for any future senolytic trial and addresses the subgroup problem (P 41025901).
4. **If, and only if, steps 1–3 are favorable — a biomarker-anchored go/no-go Phase 1b/2a.** Design: subgroup-enriched (high senescence-burden biomarker, P 41025901); intermittent "hit-and-run" navitoclax dosing to exploit the primate CSF-senolysis window while limiting thrombocytopenia (P 39253182); primary endpoints on CSF/PET SASP and neuroinflammation plus intensive platelet and neurofilament-light safety monitoring, with ALSFRS-R/survival as secondary. Given current evidence, **D&Q — not navitoclax — is the more rational first-in-ALS senolytic to carry into humans**, because it already has an ALS-model efficacy signal and a favorable human-neuron safety profile.

Verdict

The case that *cellular senescence* is a real, early, spatially appropriate, and progression-linked mechanism in ALS is **moderately strong** and supported by both mouse and human data. The case that *navitoclax* is the right drug to exploit it is **weak-to-contraindicated**: navitoclax inhibits BCL-2 and BCL-xL, which multiple independent ALS studies show are neuroprotective for motor neurons; the only validated ALS senolytic is D&Q; the one direct human-neuron comparison found navitoclax neurotoxic and D&Q safe; the accelerated-aging rationale is largely confounded; single-pathway senolysis is likely insufficient in a multistep disease; and navitoclax carries severe on-target thrombocytopenia with incomplete senolysis due to MCL-1 redundancy. The single most efficient next experiment is **BH3 apoptotic-priming profiling across ALS motor neurons, astrocytes, and microglia** to determine whether any navitoclax therapeutic window exists before any patient is exposed; absent such a window, D&Q is the more rational senolytic to advance in ALS.

Highest-weight identifiers

- **PMID 40196013** — The actual positive ALS senolytic result — but with D&Q, not navitoclax.
- **PMID 10582606** — Bcl-2 overexpression is neuroprotective in ALS → inhibiting it (navitoclax) is the wrong direction.
- **PMID 40713864** — Navitoclax toxic to human neurons in vitro where D&Q was safe.
- **PMID 42794664** — Senescence in human sporadic-ALS astrocytes, CSF and tissue, tracking progression (mechanism active in humans).
- **PMID 31905614** / **PMID 25986728** — BCL-xL is neuroprotective; raising it extends ALS survival (second contradiction).
- **PMID 42378247** — Grade 3/4 thrombocytopenia in 54% (dose-limiting on-target toxicity).
- **PMID 39253182** — Navitoclax reaches CNS at senolytic CSF levels but causes reversible thrombocytopenia (feasibility + safety).
- **PMID 39762561** — Many senescent cells resist navitoclax via MCL-1/GPX4 redundancy → senolysis often incomplete.
- **PMID 41025901** — "Accelerated aging" in ALS is largely confounded; real only in a genetic subgroup (defines who to treat).
- **PMID 41750236** — ALS multistep model: single-pathway/late senolysis likely insufficient.
- **PMID 42759848** — Senescence is an early feature of motor cortex/spinal cord in a TDP-43 ALS model (mechanism active).

- **NCT04063124** / **NCT04685590** — Only human CNS senolytic trials; D&Q in Alzheimer's, not navitoclax in ALS.

Report prepared from a 5-iteration autonomous investigation; 44 papers reviewed, 10 findings confirmed. Established findings are cited by PMID/NCT with verbatim abstract quotes; interpretive statements are explicitly labeled INFERENCE.
